

Bangladesh University of Business and Technology



LAB REPORT

Experiment No : 02
Course Title : Structured Programming Language Lab
Course Code : CSE-101
Experiment Name: Array

Submitted By:

Name : Suborna Akter
ID No : 20254103352
Intake : 56
Section : 09
Program : B.Sc. in CSE

Submitted To:

Name : Shuvo Biswas

(Lecturer)
Dept. of CSE
**Bangladesh University of Business
& Technology**

Date of Submission : 10/26/2025

Lab Report (*array*)

Problem Name:

1. Write a C program to read and print elements of array.

Source Code:

```
#include <stdio.h>

int main()
{
    int n, i;

    printf("Input size: ");

    scanf("%d", &n);

    int arr[n];

    printf("Input elements:\n");

    for (i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("\nOutput:\n");

    printf("Output: ");

    for (i = 0; i < n; i++)
    {
        printf("%d", arr[i]);

        if (i < n - 1)
        {
            printf(", ");
        }
    }

    printf("\n");

    return 0;
```

```
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS F:\Lab Report> gcc "f:\Lab Report\1_array.c" -o 1_array
e }
Input size: 10
Input elements:
1
2
3
4
5
6
7
8
9
10

Output:
Output: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
```

Problem Name:

2. Write a C program to print all negative elements in array.

Source Code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n = 10;
```

```
    int arr[n];
```

```
int i, first = 1;
printf("Input array:\n");
for (i = 0; i < n; i++)
{
    scanf("%d", &arr[i]);
}
printf("\nOutput:\n");
printf("Output: ");
for (i = 0; i < n; i++)
{
    if (arr[i] < 0)
    {
        if (!first)
        {
            printf(", ");
        }
        printf("%d", arr[i]);
        first = 0;
    }
}
printf("\n");
return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS F:\Lab Report> gcc "f:\Lab Report\2_array.c"
e }
Input array:
-1
-10
100
5
61
-2
-23
8
-90
51

Output:
Output: -1, -10, -2, -23, -90
```

Problem Name:

3. Write a C program to find sum of all array elements.

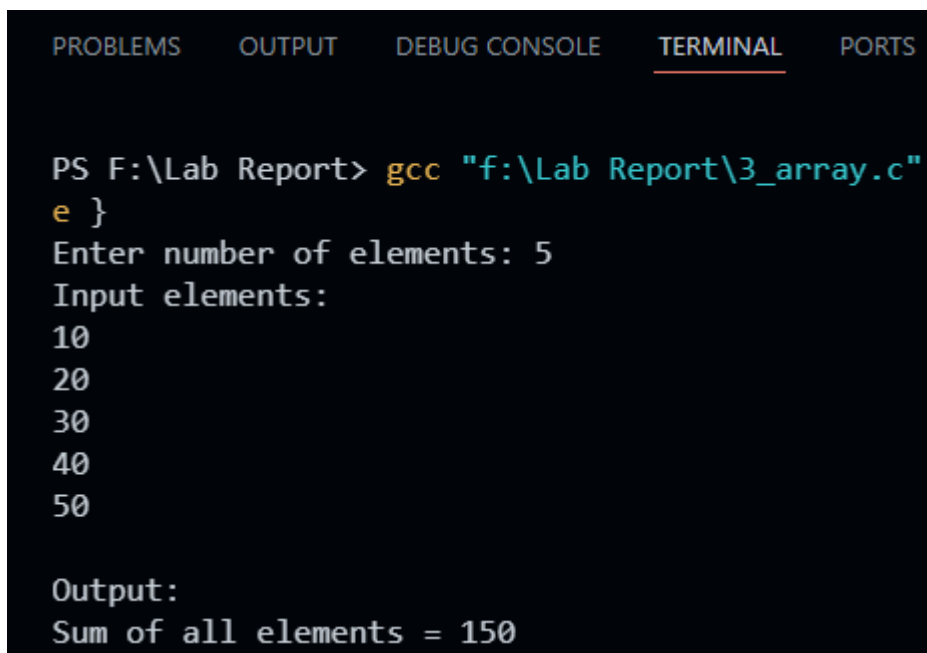
Source Code:

```
#include <stdio.h>

int main()
{
    int n, i, sum = 0;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    int arr[n];
    printf("Input elements:\n");
    for (i = 0; i < n; i++)
    {
```

```
        scanf("%d", &arr[i]);
    }
    for (i = 0; i < n; i++)
    {
        sum += arr[i];
    }
    printf("\nOutput:\n");
    printf("Sum of all elements = %d\n", sum);
    return 0;
}
```

Output:



The screenshot shows a terminal window with a dark background. At the top, there are tabs labeled 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is selected and underlined), and 'PORTS'. The terminal content shows the command prompt 'PS F:\Lab Report>' followed by the command 'gcc "f:\Lab Report\3_array.c"'. The program then prompts 'Enter number of elements: 5' and 'Input elements:'. The user enters the numbers 10, 20, 30, 40, and 50 on separate lines. Finally, the program outputs 'Output:' followed by 'Sum of all elements = 150'.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS F:\Lab Report> gcc "f:\Lab Report\3_array.c"
e }
Enter number of elements: 5
Input elements:
10
20
30
40
50

Output:
Sum of all elements = 150
```

Problem Name:

4. Write a C program to find maximum and minimum element in an array.

Source Code:

```
#include <stdio.h>

int main()
{
    int n, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Input array elements:\n");
    for (i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    int max = arr[0];
    int min = arr[0];
    for (i = 1; i < n; i++)
    {
        if (arr[i] > max)
        {
            max = arr[i];
        }
        if (arr[i] < min)
        {
            min = arr[i];
        }
    }

    printf("\nOutput:\n");
    printf("Maximum = %d\n", max);
    printf("Minimum = %d\n", min);
    return 0;
```

```
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS F:\Lab Report> gcc "f:\Lab Report\4_array.c"
e }
Enter number of elements: 5
Input array elements:
110
50
12
16
2

Output:
Maximum = 110
Minimum = 2
```

Problem Name:

4. Linear Search

Source Code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n, i, key, found = 0;
```

```
    printf("Enter number of elements: ");
```

```
    scanf("%d", &n);
```

```
    int arr[n];
```

```
    printf("Enter %d elements:\n", n);
```



```

for (i = 0; i < n; i++)
{
    scanf("%d", &arr[i]);
}

printf("Enter the value to search: ");
scanf("%d", &key);
for (i = 0; i < n; i++)
{
    if (arr[i] == key)
    {
        printf("Value %d found at position %d\n", key, i + 1);
        found = 1;
        break;
    }
}
if (found == 0)
{
    printf("Value %d not found in the array.\n", key);
}

return 0;}

```

Output:

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS F:\Suborna CSE Lab Reepor...> gcc "f:\Suborna CSE Lab Reepor...
exe -mconsole ; if ($?) { ./linear_search.exe }
Enter number of elements: 5
Enter 5 elements:
10
20
30
45
60
Enter the value to search: 45
Value 45 found at position 4

```